Diabetes Prediction Using Quantum Machine Learning

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***Abstract*—Diabetes is a chronic metabolic disorder in which high blood glucose levels are caused by a deficiency in insulin secretion or an inability to use it effectively. Diabetes is a condition that can significantly threaten health and cause serious harm if not controlled: major complications include cardiovascular disease, nerve damage, renal failure, vision impairment, reducing quality of life and imposing heavy burdens on healthcare systems. This early alert to healthcare professionals opens opportunities for intervention to help prevent and reduce complications of diabetes. Traditional ML has been applied in the prediction of diabetic risk, with exciting results based on historical patient data that indicate prevalent risk patterns. However, their performance is often limited by the intrinsic complexity of diabetes and its risk factors, such as the non-linear interactions among features like age, BMI, blood pressure, and blood glucose levels. The QML approach addresses these limitations. This paper predicts diabetes using QML techniques, taking advantage of specific properties of quantum computing, namely quantum parallelism, entanglement, and superposition-in order to process information that cannot be classically processed. The model, based on this hybrid approach with a quantum circuit for encoding relevant features, is able to capture complexity in data patterns that might be missed by classical models. It also focuses on model interpretability alongside predictions. Knowing which features (for example, blood glucose levels, HbA1c, BMI, age) contribute the most to the decision is critical in clinical use. We calculate the feature importance in the QML model that explains the most significant physiological and demographic factors associated with diabetes risk. Such insights are helpful to clinicians in formulating appropriate prevention and management plans tailored to individual risk profiles, in line with the personalized medicine approach to tailoring treatment based on the basis of risk profiles. To make it accessible, we designed a friendly and web-based user interface for healthcare professionals to input patient data, derive predictions, and interpret the outcomes. Access to the QML model is simplified so the predictive analytics can reach a wider audience. The successful integration of QML into diabetes prediction, coupled with an intuitive interface, speaks for the profound potential of quantum computing in healthcare.**

I. INTRODUCTION

Diabetes mellitus is a chronic disease; mainly that which is referred to as diabetes. It's a metabolic disorder involving an increase of blood glucose over an extended period without proper treatment causing serious health complications. These complications include cardiovascular disease, failure of kidneys, nerve damage, and loss of vision. Diabetes causes about 1.5 million deaths per year, according to WHO. The diabetes disease burden is increasing globally both in Type 1 and Type 2. Type 1 diabetes is an autoimmune condition where the body attacks cells producing insulin, and therefore, has to be managed, hence throughout life with insulin. Type 2 diabetes is the most common form of diabetes and is often associated with lifestyle factors like diet and physical activity and is associated with resistance to insulin. Effective management of diabetes depends greatly on early detection and timely intervention. That is where the importance of predictive models lies. What predictive models would do is allow for the possibility of opening up ways that may defer or even avoid the onset of the disease. Predictive models can analyze health data, such as blood glucose level, body mass index, blood pressure, and family history, that were put together in a matrix to determine the risk a particular person has at developing diabetes. However, it is challenging to precisely predict diabetes because these interactions are complex and nonlinear. The present project therefore investigates the possibility of applying Quantum Machine Learning to improve prediction accuracy in diabetes, taking advantage of the capabilities of quantum computing when dealing with complex and very high-dimensional data.

II. PROBLEM STATEMENT

The paper will focus on the development of predictive diabetes models using QML technology. This will provide a hybrid model; it integrates quantum computing and classical machine learning in utilizing the power of quantum mechanics when dealing with complexities surrounded by predictability in diabetes. This model would be developed to handle high-dimensional, nonlinear data much better than a conventional model and give meaningful insights into the risk factors influencing diabetes. This research will identify how well QML can be used for the improvement of predictive accuracy in diabetes and assess the performance of the model versus classical machine learning models. Moreover, this project shall work towards making the model user-friendly, web-based, and thus accessible for healthcare providers to integrate it in clinical workflow for early diagnosis of diabetes.

III. LITERATURE REVIEW

The quantum computing and artificial intelligence converged phenomenon is called quantum machine learning. It presents promising new avenues toward the effective management of complex medical datasets. Because QML can analyze large amounts of data with an unmatched level of computational efficiency, it is becoming an increasingly promising approach in medical diagnostics concerning diseases such as diabetes and cardiovascular disease.

Gupta et al. [1] have compared the relative performance of quantum machine learning models with respect to classical deep learning approaches in predicting diabetes. The results show that QML models can reach higher accuracy and computationally process much faster due to inherent parallel processing capacities that also make them useful for big medical data size as well. Hancco-Quispe et al. [2] further explore QML's application in a diabetes classification task, which manifests quantum algorithms' ability to deal with noisy and imbalanced datasets usually found in medical applications. From this study, the basis has been laid for the application of quantum-based classifiers in health care.

Apart from clinical predictions in diabetes, promising results are also seen in applications of quantum machine learning in other sectors of medicine. El Maouaki et al. [3] attempted to apply QSVM on the prognosis of prostate cancer; a detailed performance analysis highlighted the advantages of quantum techniques over traditional approaches in terms of speed and accuracy. Related to this is the impact of various quantum kernels by Akpinar et al. [4]. Their contributions showed that the performance of QML models is highly based on the kernel that was chosen for the said model while deployed on medical datasets; therefore, this is an area to be considered for further work.

Hybrid methods that integrate quantum computing with traditional ML have been researched. AboHashish and Algarni [5] developed a new hybrid kernel method based on support vector machines and particle swarm optimization that predicts cardiovascular disease. Their findings found that hybridization of quantum-inspired techniques with classical optimization algorithms does significantly enhance the prediction quality. Saxena [6] proposed a quantum-inspired model for neural networks on heart disease prediction, generally improving generalization and predictive accuracy in comparison to the standard neural network.

Quantum machine learning is an area of recent interest due to the fact that applications in medicine strictly require interpretability. A possible explainable ensemble-quantum machine learning approach towards heart disease prediction is suggested by Shaiba [12]. This model brought practical insights into features and helped clinicians understand the reasons behind the prediction. Alotaibi [7] explains scalability and effectiveness in quantum-enhanced algorithms as the reason for the power of the algorithm in handling complex data related to clinical problems, which further increases predictive accuracy.

Flöther et al. [14] illustrated how quantum technologies would find a possible foothold in healthcare, elaborating applications of quantum computing in health and medicine; the research on the subject propounds the capability of quantum computing in the shaping of diagnostic tools, treatment methods, and personalized medicine. Moreover, Maheshwari et al. [15] have compared quantum and classical computation methods used in the diabetes prediction and concluded that the quantum systems hold promise for faster and accurate prediction speeds and overcome the shortcomings of the cla classical models

The package of these studies collectively speaks to the transformative potential of QML into healthcare. Quantum models seem to explore what otherwise are subtle and complex relationships within the data: by quantum mechanical principles of superposition and entanglement. Where other classical algorithms, though theoretically calculable, are just too computationally intensive to run, advances in QML promise diabetes prediction and myriad medical diagnostics as important strides toward the realization of precision medicine wherein not only are outcomes predicted but results are interpretable and actionable.

IV. METHODOLOGY

The work uses a hybrid methodology that combines traditional and quantum machine learning for conditions of diabetes. Each phase, including preprocessing, model development, training, evaluation, and deployment, represents the kinds of strategies that might at least address the various forms of complications related to healthcare data. A baseline is a classical Support Vector Machine model, and QSVM implements improved quantum methods of overcoming the traditional limitations of ML. The methodology outlined is sufficiently explicit, covering data handling, architectures of models, and evaluation metrics needed for achieving results that are potentially accurate but still interpretable.

A. Data Preprocessing

Data preprocessing included handling missing values and scaling of data, encoding categorical variables, and SMOTE to maintain a balance of classes. PCA performed optimization of data to train the QSVM model. Reduction of dimension with PCA is relevant because appropriate processing of the data is quintessential to the QSVM model. The robust benchmark for SVM model was developed, and its hyperparameters were tuned by grid search while QSVM model resulted in representation with entanglement and also quantum kernel mapping due to its capability of presenting complex patterns of data.

B. Exploratory Data Analysis

A preliminary exploratory analysis of the features was done to get an overall understanding of the data and for any potential inconsistencies or patterns that it may have. The dataset contains demographic characteristics, like gender and age; physiologic measures such as BMI, HbA1c, blood glucose, among others, and health history markers that include heart disease, hypertension, and smoking history. From the distribution plots we observe:

The age distribution shows most patients with diabetes are above 55 years of age, hence establishing a strong relationship between advancement of age and the risk of diabetes.

Values of HbA1c above 7 are also important cut-off points, since anyone with HbA1c levels above this point is now classified as diabetic. This resonates with received opinion in medicine-which posits that a HbA1c above 6.5 points to

inadequately controlled glycemia.

Non-diabetic subjects have a more uniform distribution with regards to different age groups and HbA1c levels below 7; those two features are very important to predict diabetes. The results obtained informed the ensuing preprocessing procedures, including scaling and outlier identification, thereby guaranteeing that the models proficiently recognized these patterns for precise forecasting.

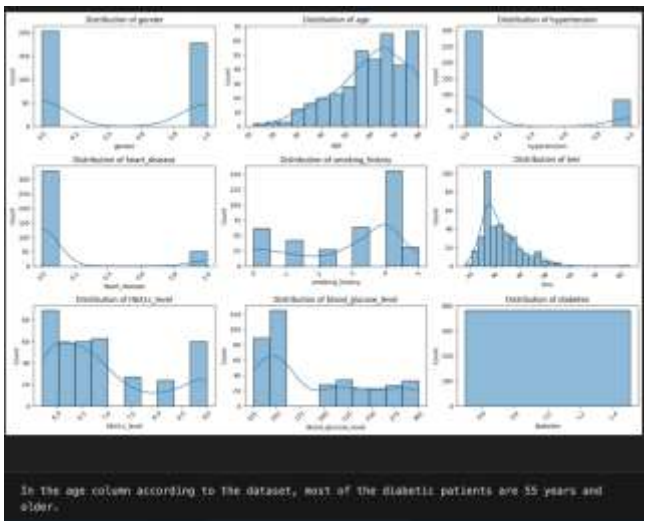

Fig. 1. Distribution Plot for Diabetes True

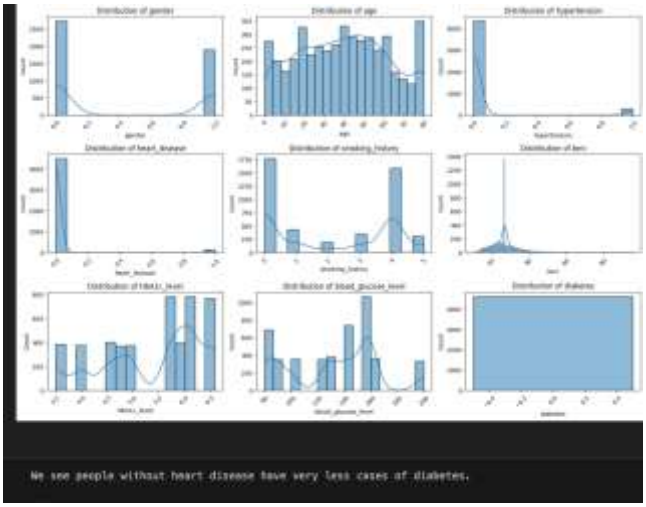

Fig. 2 . Distribution Plot for Diabetes True

C. Model Development

The research established two separate models for diabetes prediction, a conventional Support Vector Machine SVM and Quantum Support Vector Machine QSVM, both of which showed some strength during the analysis of diabetes data.

The classical SVM model was selected as a baseline due to its established reliability in binary classification tasks.

• Kernel Selection: The Radial Basis Function kernel was applied in Because of its efficiency in capturing relationships between nonlinear variables

Hyperparameter Tuning:

a. Penalty Parameter C: Adjusted to control the amount of the margin by smoothing the trade-off between a greater margin and superior classification performance.

b. Gamma: The impact of individual training examples specified inside the paper; how smaller values of gamma include all the more general patterns and higher values focus on closer neighbors.

A QSVM model incorporates quantum computation for input features transformation into a high-dimensional quantum state space that may contribute to the enhancing ability of the model in complex data pattern identification. Here 6 qubits are used.

• Quantum Kernel Definition: It converts classical data into quantum states, making it possible to execute the quantum kernel function. The AngleEmbedding method from PennyLane is used to encode every feature as a rotation angle in a quantum circuit.

• Quantum Circuit Design:

Angle encoding: Every feature value is encoded as an angle for which the qubits rotate. In this method, the quantum state encodes the feature value.

Entanglement with CNOT Gates: Quantum gates entangle qubits with each other, signifying intricate relationships between features.

• Kernel Matrix Computation: Apply the quantum kernel to every pair of the training data and generate the quantum kernel matrix. Input this into the SVM classifier, which captures very complex interrelations.

D. Evaluation Metrics

Accuracy, precision, recall, and F1-score will be used for assessing comparative performance of both the models. These metrics will provide in-depth insights into which of the models can accurately predict the disease as well as their viability in practice-oriented healthcare provisions.

• Accuracy: Accuracy gives the proportion of instances correctly predicted by the model out of the total instances. However it does not measure precisely the performance of an actual model; this may happen in a scenario where class distributions are uneven, which is a real frequent case for health data.

Formula: Accuracy = (True Positives + True Negatives) / (Total Predictions)

Interpretation: A high accuracy score may indicate that this model classifies diabetic cases and non-diabetic cases close to the actual. Other metrics, however, have to be considered so as to effectively appreciate this model's performance.

Here for our model, SVM has an accuracy of 89.7% and QSVM has an accuracy of 88.7%.

• Precision: Precision is the ratio of true positive predictions

to all positive predictions. It basically indicates a good model that can avoid false positives because unnecessary treatment or intervention may be used if a patient develops false positivity in healthcare settings.

Formula: Precision = True Positives / (True Positives + False Positives)

Interpretation: A high accuracy level translates to the reliability that this model is able for the right identification of actual diabetic patients without over-prediction of the disease among non-diabetic populations.

Here for our model, SVM has a precision of 91% and QSVM has a precision of 89%.

Recall: Recall measures a model's sensitivity. Specifically, recall is the ratio of true positives to the total actual positives. While predicting diabetes, high recall is needed so that as many cases of diabetes as possible can be identified.

Formula: Recall = True Positives / (True Positives + False Negatives)

Interpretation: A model characterized by high recall reduces the chances of missing cases of diabetic patients; therefore, there is an increased chance that the diagnosed patients get earlier intervention.

Here for our model, SVM has a recall of 88% and QSVM has a recall of 88%.

F1 Score: The F1 score, thus, is basically the harmonic mean of precision and recall and that whenever a balanced approach between two metrics is desired, this should be utilized. This is very important in healthcare settings due to the need for a balance between minimizing false positives and maximizing true positives.

Formula: F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

Interpretation: A high F1 score means that a model is balanced between precision and recall and, therefore, more likely to be clinical where both of these metrics are of utmost value.

Here for our model, SVM has an F1 score of 90% and QSVM has an F1 score of 88%.

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Accuracy: 0.887987012987013

Classification Report:
              precision    recall  f1-score   suppo

           0       0.89      0.88      0.88       90
           1       0.89      0.90      0.89       94

    accuracy                           0.89      184
   macro avg       0.89      0.89      0.89      184
weighted avg       0.89      0.89      0.89      184
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Fig 3.Classification Report for QVSM Model

E. Web Based User Interface for Prediction

The QSVM model was saved in a quantum-compatible format using a custom function within the quantum library. The saved model, with the kernel matrix for the quantum kernel, allowed for the making of predictions without the kernel having to be recomputed at each session. A web interface has been developed to make the system accessible to the health practitioners. Here users are able to input patient data, obtain predictions from the model, and see associated risk scores.

Frontend: An intuitive interface to input data and present results has been developed with HTML, CSS, and JavaScript.

• HTML: HTML is, naturally, the language in which this form structure relies on semantic tags for structuring in a coherent sequence, thereby improving accessibility for screen readers.

• CSS Cascading Style Sheets: CSS is one style type, such as color schemes, font selections, and spacing. It ensures consistency in the style. In other words, it improves the aesthetic appeal of the interface and its usability.

• JavaScript: JavaScript oversees the interactivity and the dynamic behavior of the interface. It does real-time validation for user inputs, submits a form and lets off AJAX requests, so the UI remains within contact with the backend without reloading a page.

Backend: The frontend with stored models was thus linked by Flask, making real-time predictions possible.

• Flask: It is a Python-based microframework that can take care of HTTP requests and backend logic management. Flask is naturally ideal for the rapid development for the integration of Python-based machine learning models.

• Importing the Model and Application: The classical SVM model and the QSVM will be imported within the Flask application. Routes of application will manage the prediction task; this involves receiving POST requests, applying the model, and returning the result.

• API Endpoint: A RESTful API endpoint has been defined to receive requests for prediction. The frontend sends a POST request to this endpoint containing the form data, and the endpoint sends its response in the form of a JSON prediction.

V. CONCLUSION

This study explored the ability of Quantum Machine Learning (QML) in diabetes prediction through the development and comparison of a classical Support Vector Machine (SVM) model with a Quantum Support Vector Machine (QSVM) model. Through rigorous data preprocessing, training, and evaluation, the following critical observations were derived, Accuracy-wise, the classical SVM model had a higher accuracy since it was able to attain 89.7%, compared to QSVM, which only attained 88.7%. The classical model performed very well in terms of precision, recall, and F1 score and can therefore be relied on to predict diabetes. Although QSVM was not stronger in terms of accuracy in comparison to classical SVM, it showed that quantum kernels can identify complex patterns in data.

QML entails hardware limitations, challenges with regard to interpretability, and computational cost as some of the several key challenges that its applications in healthcare come to face. All these limitations underpin the continued development of quantum computing and algorithms. QSVMs are still in the early stages of practical application and excel in tasks where classical approaches struggle with high-dimensional quantum feature spaces. However, for many tasks, a classical SVM remains a strong and reliable model, especially for moderately sized datasets

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